

Assignment:

Assuming the table is named **tfi** and it has these columns:

- emp_name
- dept
- salary

Create a CTE to find the average salary of each department.

```
WITH dept_avg AS (  
  SELECT dept, AVG(salary) AS avg_salary  
  FROM tfi  
  GROUP BY dept  
)  
SELECT *  
FROM dept_avg;
```

Create a CTE to find the maximum salary of each department.

```
WITH dept_max AS (  
  SELECT dept, MAX(salary) AS max_salary  
  FROM tfi  
  GROUP BY dept  
)  
SELECT *  
FROM dept_max;
```

-- Create a CTE that calculates the average salary for each department. Then display
only departments

-- whose average salary is greater than 70000

```
WITH dept_avg AS (  
  SELECT dept, AVG(salary) AS avg_salary  
  FROM tfi  
  GROUP BY dept  
)  
SELECT *  
FROM dept_avg  
WHERE avg_salary > 70000;
```

```
SELECT dept, AVG(salary) AS avg_salary
FROM tfi
GROUP BY dept
)
SELECT *
FROM dept_avg
WHERE avg_salary > 70000;
```

-- Using a CTE, find departments having more than 5 employees.

```
WITH dept_emp_count AS (
  SELECT dept, COUNT(*) AS employee_count
  FROM tfi
  GROUP BY dept
)
SELECT *
FROM dept_emp_count
WHERE employee_count > 5;
```

-- Create a CTE containing the total salary paid by each department. Display the department with

-- a total salary greater than 500000

```
WITH dept_total_salary AS (
  SELECT dept, SUM(salary) AS total_salary
  FROM tfi
  GROUP BY dept
)
SELECT *
FROM dept_total_salary
WHERE total_salary > 500000;
```

-- Using a CTE, find the highest salary in each department.

```
WITH dept_highest_salary AS (  
    SELECT dept, MAX(salary) AS highest_salary  
    FROM tfi  
    GROUP BY dept  
)  
SELECT *  
FROM dept_highest_salary;
```

-- Create a CTE containing the average salary of each department. Then join it with tfi and display:

-- emp_name, dept, salary, avg_salary

```
WITH dept_avg AS (  
    SELECT dept, AVG(salary) AS avg_salary  
    FROM tfi  
    GROUP BY dept  
)  
SELECT  
    t.emp_name,  
    t.dept,  
    t.salary,  
    d.avg_salary  
FROM tfi t  
JOIN dept_avg d  
ON t.dept = d.dept;
```

-- Find only the employees whose salary is greater than their department's average salary.

```
WITH dept_avg AS (  
    SELECT dept, AVG(salary) AS avg_salary  
    FROM tfi  
    GROUP BY dept
```

```
)  
SELECT  
    t.emp_name,  
    t.dept,  
    t.salary  
FROM tfi t  
JOIN dept_avg d  
ON t.dept = d.dept  
WHERE t.salary > d.avg_salary;
```

-- Using a CTE, find employees whose salary is LESS than their department's average salary

```
WITH dept_avg AS (  
    SELECT dept, AVG(salary) AS avg_salary  
    FROM tfi  
    GROUP BY dept  
)  
SELECT  
    t.emp_name,  
    t.dept,  
    t.salary  
FROM tfi t  
JOIN dept_avg d  
ON t.dept = d.dept  
WHERE t.salary < d.avg_salary;
```